

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

STRUCTURE OF ATOM

LECTURE-4

CHEMISTRY

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Topics To Covered

→ ISOTOPES & ISOBAR ✓

1 BOHR MODEL OF ATOM ✓ (Bahut Imp).

REVISION.

Thompson model of atom.

★ Rutherford model of atom.

- Experiment
- Observations
- Conclusion
- Postulates.

Limitations.

★ Atomic No.
+
Mass No.

(Z) + (A).

ISOTOPES + ISOBAR.

- Atoms of the same elements having same atomic number but different mass numbers are called isotopes. *e.g. Isotopes of hydrogen.* ^1_1H , ^2_1H , ^3_1H

Atoms of different elements which have same mass numbers (and of course different atomic numbers) are called isobars

e.g. $^{40}_{18}\text{Ar}$, $^{40}_{19}\text{K}$, $^{40}_{20}\text{Ca}$.

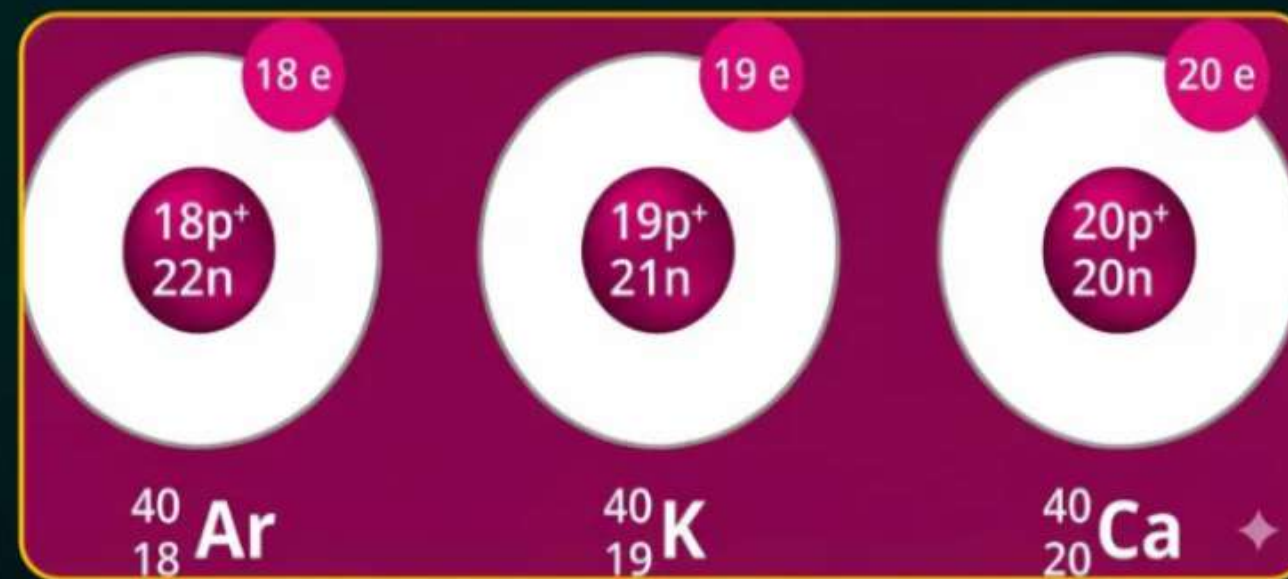
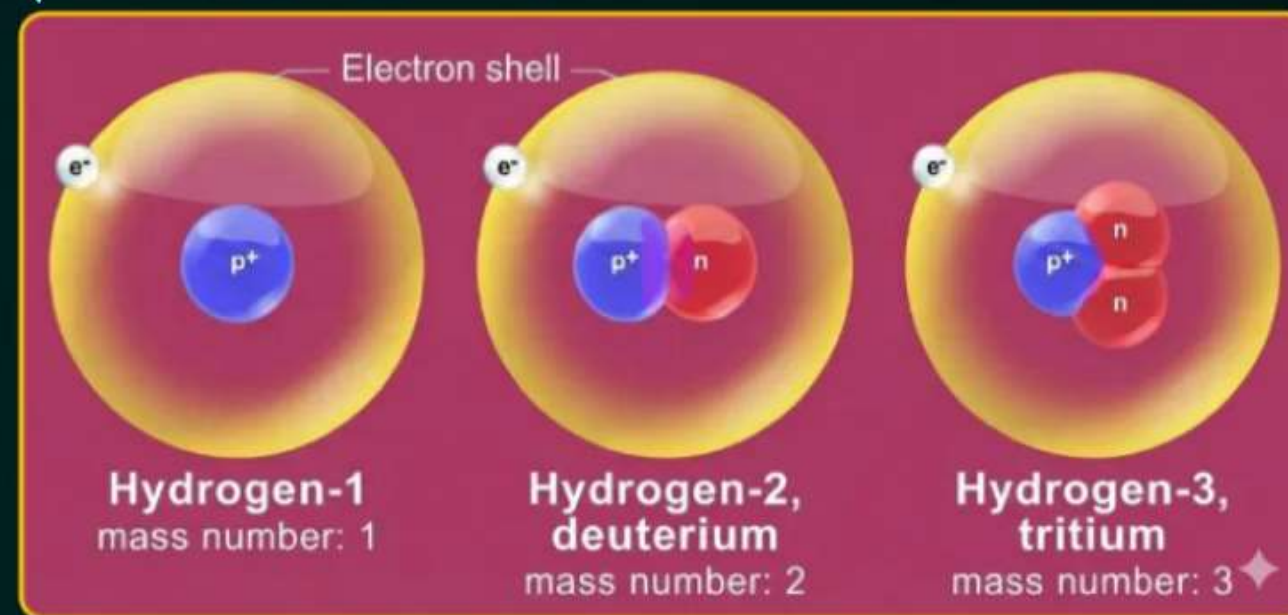
- Atoms of different element which contain the same number of neutrons are called isotones

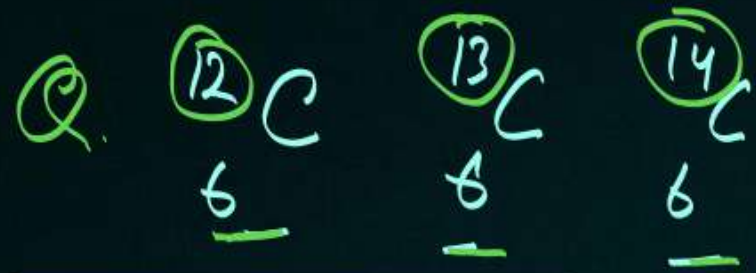
e.g. $^{14}_6\text{C}$, $^{15}_7\text{N}$, $^{16}_8\text{O}$.

$$\begin{aligned} &^{14}_6\text{C}, \quad A = P + N \\ &\therefore N = A - P \\ &= 14 - 6 \\ &= 8 \text{ neutrons} \end{aligned}$$

$$\begin{aligned} &^{15}_7\text{N}, \quad \text{Neutron} = A - P \\ &= 15 - 7 \\ &= 8 \end{aligned}$$

$^1_1\text{H} = \text{Hydrogen}$, $^2_1\text{H} = \text{Deuterium}$, $^3_1\text{H} = \text{Tritium}$.





a) Isotopes

b) Isobars

c) Isotones.

BOHR MODEL OF ATOM



To overcome the limitations of Rutherford model of atom, Neil Bohr proposed a Bohr model of atom. Important Postulates are given below $\Rightarrow$.

1) Atom consists of a heavy, positively charged nucleus present at centre and e^- revolve around it in a fixed path called orbit.



2) Out of many possible orbit e^- is present in only those orbit which have fixed energy. Here orbits are named as energy level or stationary states. Designated as 1, 2, 3, 4, 5, OR K, L, M, N, O, starting from the nucleus.

• Energy of the stationary state or energy level for H -atom or one e^- species can be calculated as. e.g. He^+ , Li^{2+} etc.

$$E_n = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J/atom.}$$

$$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV/atom.}$$

$$E_n = -1312 \frac{Z^2}{n^2} \text{ kJ/mol.}$$

BOHR MODEL OF ATOM



Order of Energy for various energy level.

$$\underline{E_1} < \underline{E_2} < \underline{E_3} < \underline{E_4} < \underline{E_5} \dots \dots \dots$$

3) Energy of stationary state or $\bar{e}$ is said to be quantised. i.e fixed energy.

4) Like energy, Angular momentum of $\bar{e}$ is also quantised.

$$mvr = n \frac{h}{2\pi}$$

5) Radius of orbit for H-atom or one $\bar{e}$ species is determined by.

$$r_n = a_0 \frac{n^2}{Z}$$

$$a_0 = 52.9 \text{ pm or } 0.52 \text{ \AA}$$

For first orbit of hydrogen atom

$$r_1 = a_0 \times \frac{1^2}{1} = a_0 = 52.9 \text{ pm}$$

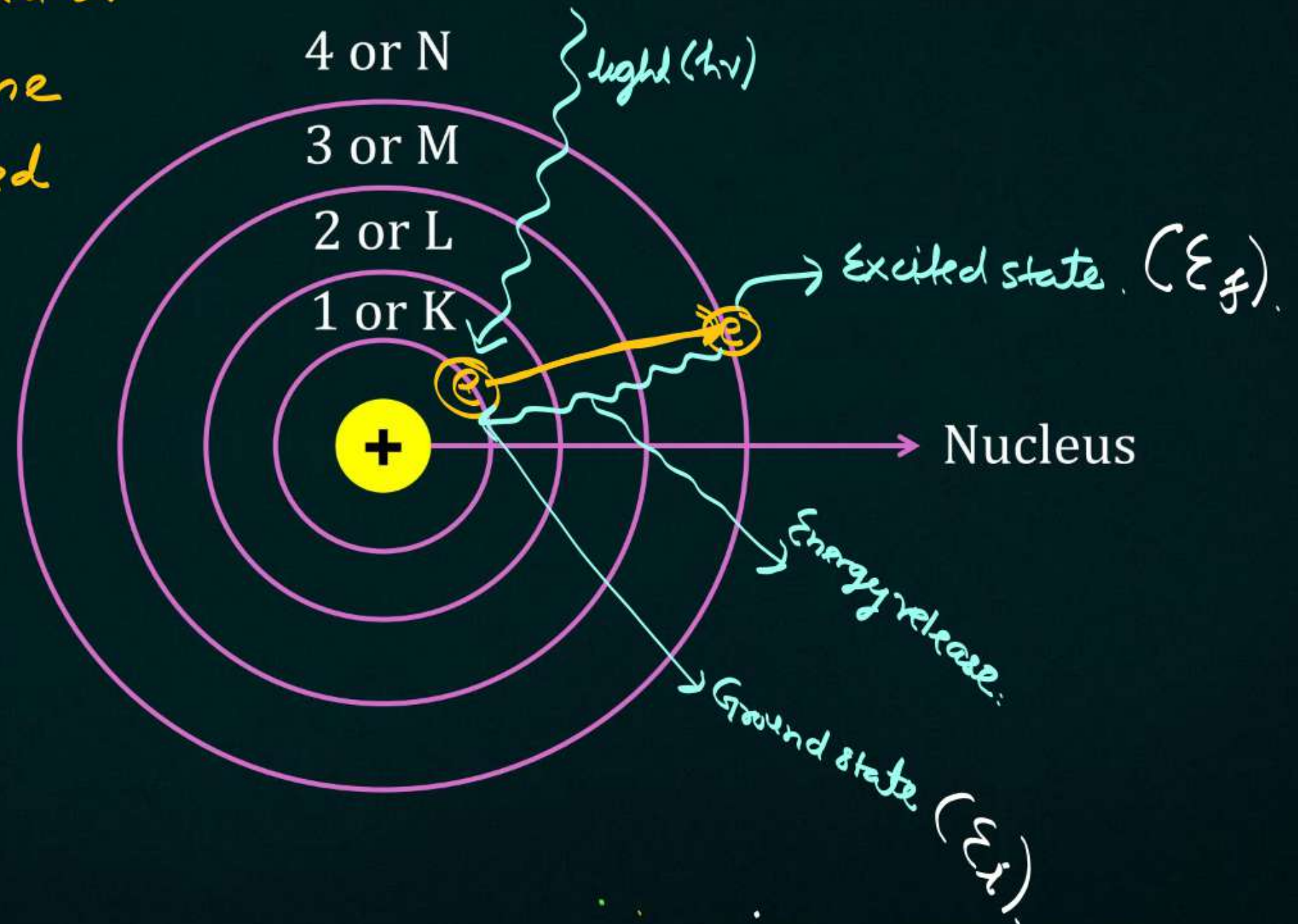
therefore first orbit radius = 52.9 pm (Bohr radius).

BOHR MODEL OF ATOM



6) Electron can not absorb or release energy continuously. Energy is absorbed or released only when e^- jumps from one orbit to another. When energy is absorbed by e^- it gets excited then releases the absorbed energy instantly & returns to the Ground state & change in energy is also calculated

$$\Delta E = E_f - E_i$$



BOHR MODEL OF ATOM



→ Centre Nucleus & e^- revolve.

→ So many orbits possible
Order of Energy

$$E_1 < E_2 < E_3 < E_4 \dots$$

→ Energy quantised (fixed)

→ Angular momentum of e^-

$$mvr = n \frac{h}{2\pi}$$

$$L_n = \frac{a_0 n^2}{Z}$$

→



$$E_n = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J/atom}$$

$$E_n = -13.6 \frac{Z^2}{n^2} \text{ eV/atom}$$

$$E_n = -1312 \frac{Z^2}{n^2} \text{ kJ/mol}$$

QUESTION



Calculate the energy associated with the first orbit of He^+ . What is the radius of this orbit?

$$Z=2$$

COMMUNITY WALA HOMEWORK

$$\begin{aligned} \underline{a_0} &= \underline{52.9 \text{ pm}} \\ &= \underline{0.52 \text{ \AA}} \end{aligned}$$

QUESTION



- (i) The energy associated with the first orbit in the hydrogen atom is -2.17×10^{-18} J atom⁻¹. What is the energy associated with the fifth orbit?
- (ii) Calculate the radius of Bohr's fifth orbit for hydrogen atom.

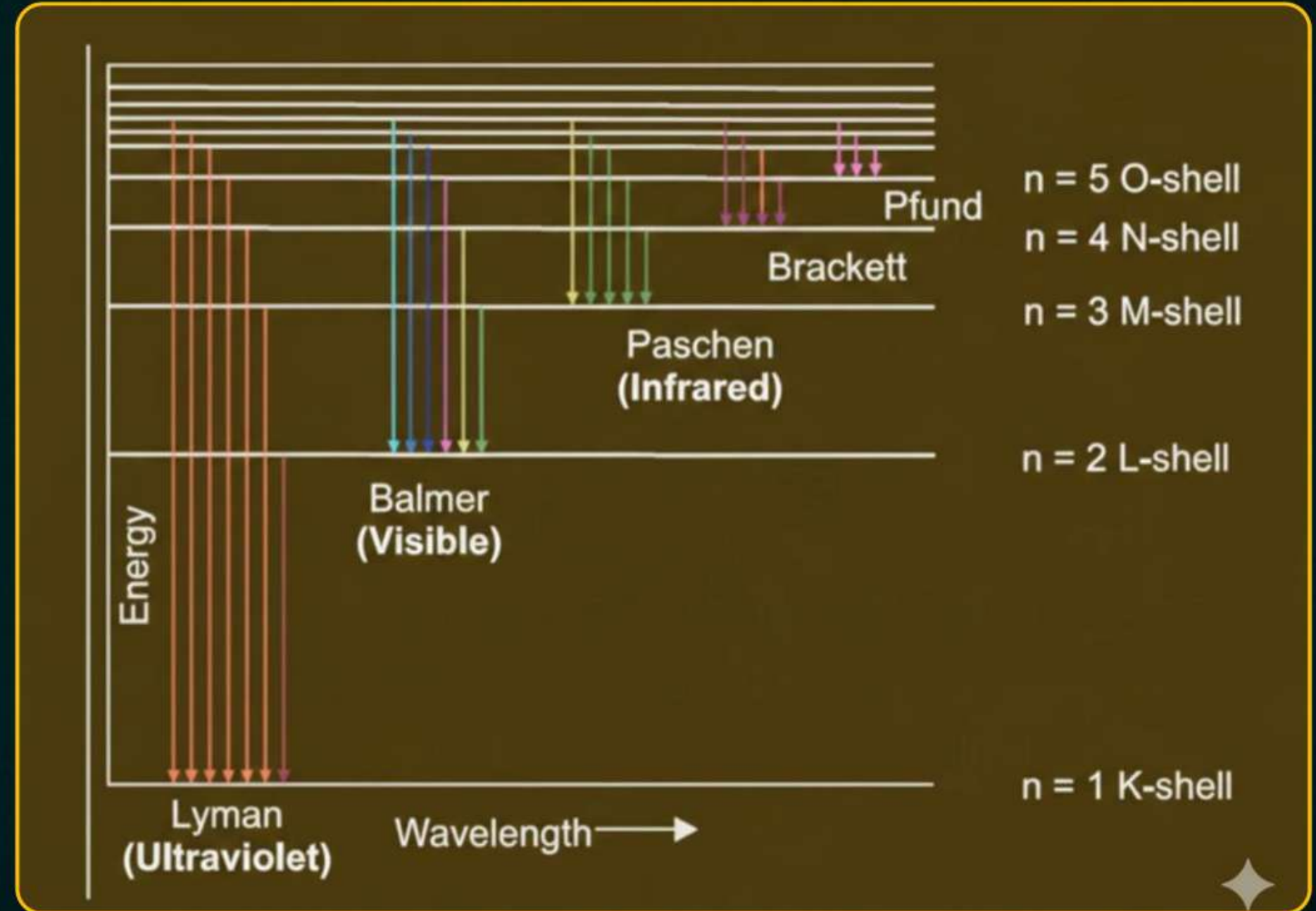
i) Sol $\rightarrow E_5 = -2.17 \times 10^{-18} \times \frac{Z^2}{n^2}$ J/atom

$$= -2.17 \times 10^{-18} \times \frac{1^2}{5^2}$$
$$= -2.17 \times 10^{-18} \times \frac{1}{25} \text{ J/atom.}$$

$$r_5 = a_0 \frac{n^2}{Z} = a_0 \frac{5^2}{1}$$
$$= 25 a_0$$

Line Spectrum of Hydrogen

- Heating or electric discharge splits hydrogen molecules into atoms.
- Electrons absorb varying amounts of energy and move to higher levels (e.g., L, M, N, O).
- The excited state of electrons is short-lived.
- Electrons return to lower levels or the ground state in one or more jumps.



QUESTION



Which of the following is a limitation of Bohr's model of the atom?

- 1 It explains the spectra of hydrogen-like ions.
- 2 It does not explain the splitting of spectral lines.
- 3 It accounts for the doublet fine structure.
- 4 None of these.

QUESTION

According to Bohr's model, the energy associated with the first orbit of He^+ is:

- 1 -13.6 eV
- 2 -54.4 eV
- 3 -108.8 eV
- 4 -27.2 eV

QUESTION

$$n=2$$

Calculate the radius of the second Bohr orbit for a hydrogen atom.

Given $r_n = n^2 \times 0.529 \text{ \AA}$.

$$\begin{aligned} r_n &= n^2 \times 0.529 \text{ \AA} \\ &= 2^2 \times 0.529 \text{ \AA} \\ &= 4 \times 0.529 \text{ \AA} \end{aligned}$$

1 $0.529 \text{ \AA}$

2 $1.06 \text{ \AA}$

3 $2.116 \text{ \AA}$

4 $4.232 \text{ \AA}$



QUESTION



Determine the energy required to excite an electron from the first to the third Bohr orbit in a hydrogen atom.

1 10.2 eV

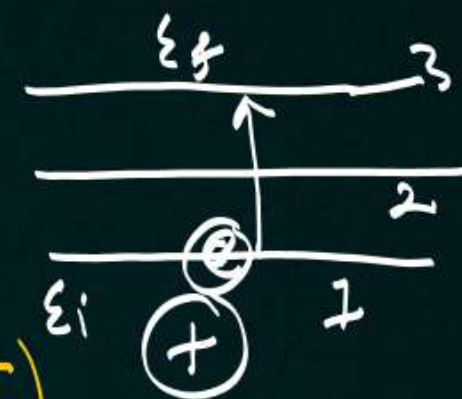
2 12.09 eV

3 13.6 eV

4 14.2 eV

$$\begin{aligned}\epsilon_3 &= -13.6 \times \frac{Z^2}{n^2} \text{ eV/atom} \\ &= -13.6 \times \frac{1^2}{3^2} \\ &= -\frac{13.6}{9}\end{aligned}$$

$$\begin{aligned}\Delta E &= \epsilon_f - \epsilon_i \\ &= \epsilon_3 - \epsilon_1 \\ &= \left(-13.6 \times \frac{1^2}{3^2} \right) - \left(-13.6 \times \frac{1^2}{1^2} \right) \\ &= -\frac{13.6}{9} + 13.6 \\ &= -1.51 + 13.6 \\ &= 12.09\end{aligned}$$



QUESTION

Calculate the angular momentum of an electron in the third orbit of a hydrogen atom.

1 $\frac{h}{2\pi}$

2 $2 \times \frac{h}{2\pi}$

3 $3 \times \frac{h}{2\pi}$

4 $4 \times \frac{h}{2\pi}$

$$mvr = n \frac{h}{2\pi}$$
$$= 3 \times \frac{h}{2\pi}$$

NOTE $\Rightarrow$ PROMISE KARO

REVISION ko Aadat Bana lo. ✓

9 $\rightarrow$ 11

3 classes

Thank
You